

- Column j_1 above Row i_1 is covered as follows:
- A Dominance covers $(2, j_1)$ and $(3, j_1)$.
- Another Dominance covers $(4, j_1)$ and $(5, j_1)$.
- This process is continued until a dominance covers $(i-2, j_1)$ and $(i-1, j_1)$.
- For each column i where $j_1 < i < j_2$, The Dominances are used to cover them as follows:

- 1) A Dominance covers $(2, i)$ and $(3, i)$
- 2) A Dominance covers $(4, i)$ and $(5, i)$
- 3) A Dominance covers $(6, i)$ and $(7, i)$

- or
- For each column i where $1 < i < j_1$ and $j_2 < i < 8$, The Dominances are used to cover them as follows:

- 1) A Dominance covers $(1, i)$ and $(2, i)$
- 2) A Dominance covers $(3, i)$ and $(4, i)$
- 3) A Dominance covers $(5, i)$ and $(6, i)$
- 4) A Dominance covers $(7, i)$ and $(8, i)$

[S] Case 2: j_1 and j_2 have opposite parities. Taking into consideration the way the chessboard is covered, Row i_1 Column j_2 is Black. $\therefore (j_2 - i_1)$ must be even.

By Defn. 1, $\exists$ integer x st. $(j_2 - i_1) = 2x$.

Let, i_1 be even. $\therefore \exists$ integer y st. $i_1 = 2y$

By Defn. 1, i_2 is even.

Let, i_1 be odd. $\therefore \exists$ integer y st. $i_1 = 2y + 1$

By Defn. 1, i_2 is odd.

$\therefore i_1$ and i_2 must have the same parities

Let, both i_1 and i_2 be even.

$\therefore 8 - i_1 = 2 \times 4 - 2y = 2(4 - y)$. By Defn. 1, $(8 - i_1)$ is even.

$\therefore 8 - i_2 = 2 \times 4 - 2(y + 1) = 2(4 - y - 1)$. By Defn. 1, $(8 - i_2)$ is even.

Column j_1 below Row i_1 is covered as follows: